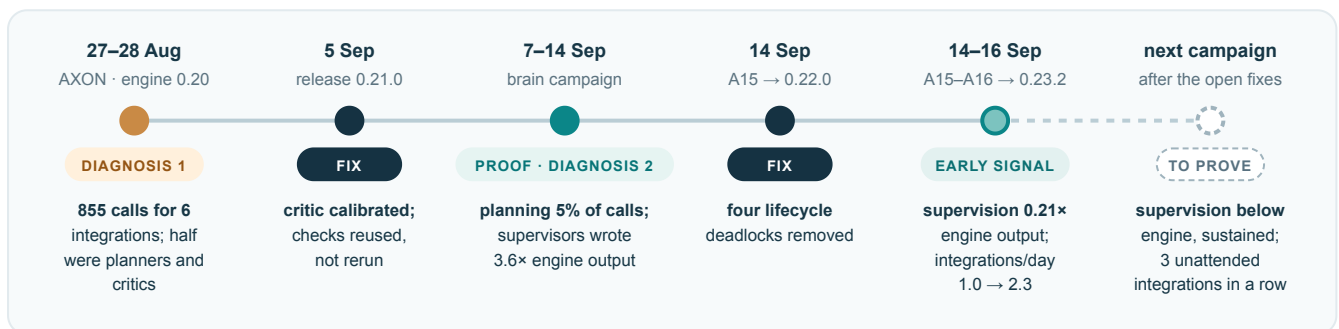


01 / SUMMARY

Measure. Fix. Measure again.

We measured two field campaigns, three weeks apart, one model call at a time. The first showed the engine arguing with itself, and that fix is proven. The second showed the cost had moved to supervising the engine from outside. Those fixes shipped in 0.22 and 0.23, and the first signal is strong. One clean run still has to prove it.



142 → 17

engine calls per integration, August to September.

949k → 44k

supervisor output tokens per integration, before and after the lifecycle fixes went live.

33% → 60%

share of engine output that became code on the integration branch, over the same split.

FIXED · PROVEN

In-loop ceremony. Planners and critics went from half of all calls in August to 8 of 166 in September.

FIXED · EARLY SIGNAL

Lifecycle deadlocks. Four removed in 0.22.0, plus two supporting fixes in 0.23. Supervision per integration fell 22×. The result is not yet controlled.

OPEN · FIX PROPOSED

Five items. Two lifecycle deadlocks, the auditor's first-try defect, retries into provider limits, worker clock sizing, and a supervision metric inside the engine.

HOW TO READ THIS

Both campaigns are field runs, not controlled experiments. The September campaign restarted 19 times on successive frozen runtimes and switched supervisor tools on 13–14 September. Codex ran on quota, so only Claude calls carry dollar amounts.

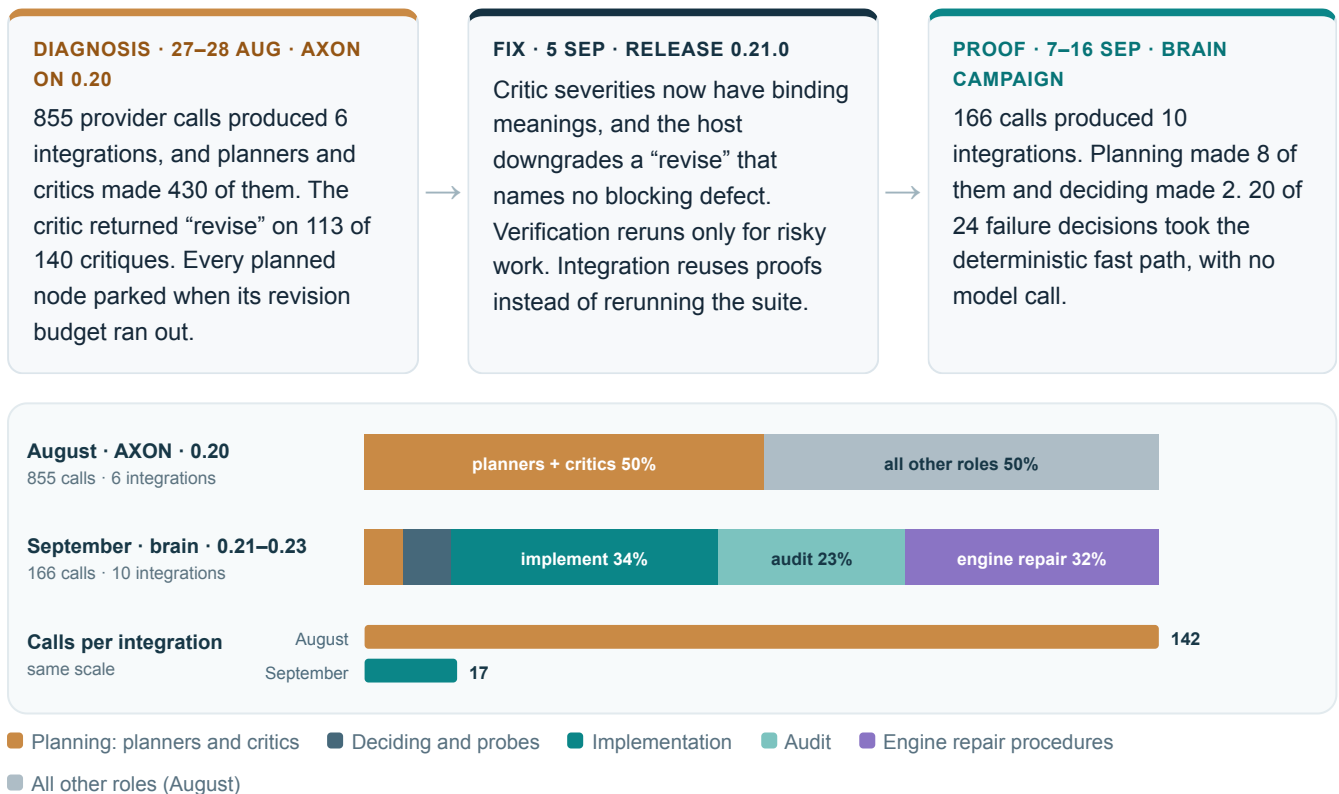
HOW TO CHECK IT

Scripts committed next to this report compute every figure from the engine's event log, git history and provider session logs. Supervisor figures come from local session stores. The August counts are in the published 0.21.0 release notes.

02 / DIAGNOSIS ONE · AUGUST

The loop argued with itself. 0.21 made it stop.

In August the engine spent most of its calls deciding what to build. After 0.21.0, planning fell from half of all calls to one in twenty. Each integration took 17 calls instead of 142.



50% → 5%

planning’s share of engine calls, August to September.

140 → 2

plan critiques. The August critic returned “revise” on 113 of its 140.

1.6%

of September engine output tokens went to planning, deciding and probes.

WHAT THE COMPARISON SHOWS

A different project, task set and campaign length, so this shows a changed role mix, not a controlled result. The three measures above are exactly the behaviours 0.21 set out to remove.

ONE LABEL TO READ CAREFULLY

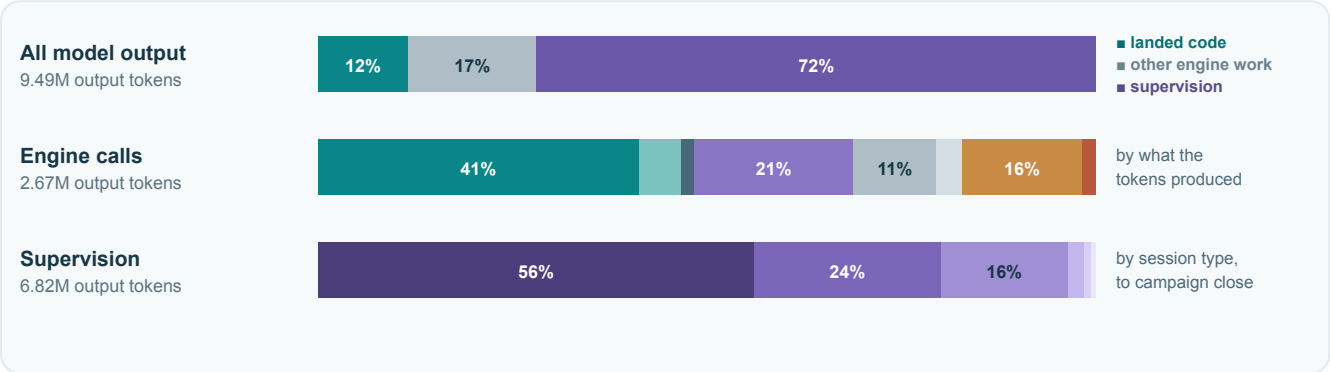
By role label, September planners and critics made 41 calls. 33 of those were engine-repair procedures a supervisor launched under the planner role. They are counted separately here, and they are the first sign of diagnosis two.

Sources. August counts come from the 0.21.0 release notes; the AXON event log independently records 6 integrations and 120 planner failures. September counts come from the campaign event log, with every `runner.completed` event counted.

03 / DIAGNOSIS TWO · SEPTEMBER

Then the overhead moved outside the loop.

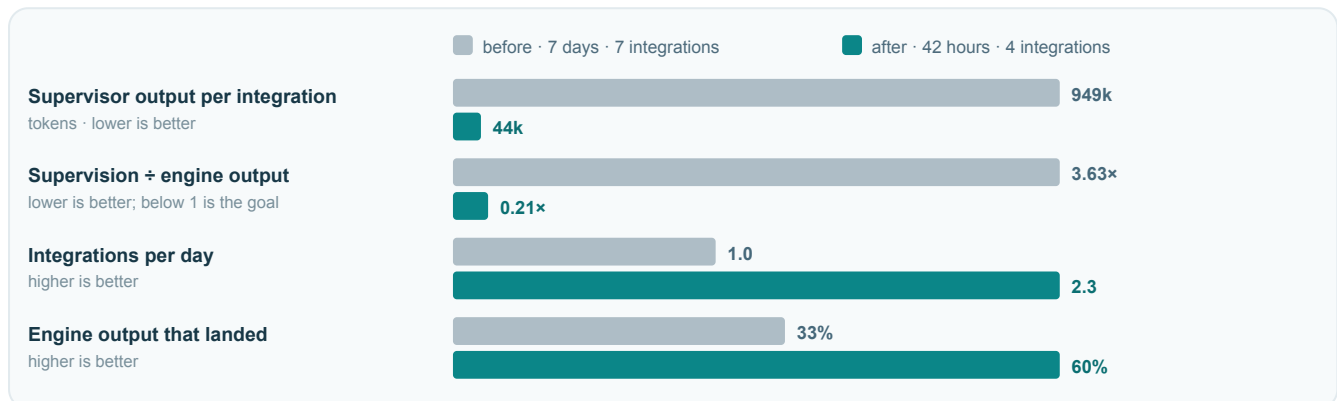
With planning fixed, the September measurement found a new cost centre. Supervisors wrote 2.6 times the engine’s output to keep it moving. Across the campaign, only 12% of all output became code on the integration branch.



04 / FIXES AND PROGRESS

After the lifecycle fixes, supervision fell 22×. One clean run must prove why.

The four lifecycle fixes went live at 16:34Z on 14 September. In the 42 hours that followed, integrations per day more than doubled. Supervisor output per integration fell from 949k to 44k tokens.



Why this is a signal, not yet proof. The supervisor tool changed on 13–14 September, from Codex with 23 subagents to a single Claude session. The earlier week includes one-off rescue and planning work that the later window did not repeat. The later window covers 42 hours and 4 integrations, and it still needed hands: the supervisor finalized one dispatch record, adopted the TASK-1114 candidate and merged TASK-1115.

FINDING	FIX: SHIPPED OR PROPOSED	STATUS
Planner and critic ceremony	0.21.0	Fixed · proven
Four lifecycle deadlocks: unpark, lease deletion, base drift, format park	0.22.0	Fixed · early signal
Complete work parked for one missing packet field	0.23.0	Fixed · not yet exercised
Integrator refusing work over a per-run setting	0.23.2	Fixed · not yet exercised
Two remaining lifecycle deadlocks	finalize on frozen exit; accept own reservation	Open
Auditor's first try fails on one field	prompt change	Open
Driver retries straight into a provider session limit	scheduler backs off; driver does not	Open
Worker clock not sized from the task	contract-sized clock	Open
Supervision invisible to engine metrics	build this measurement into the engine	Open

What will count as proof. One campaign on a release that carries the open fixes, reported per integration. It must meet three bars. Supervisor output stays below engine output across the run. Three consecutive representative integrations need no supervisor action: the repository's own qualification bar, currently 0 of 3. And at least 60% of engine output lands, the level seen after the fixes.